A.2.5. SPARSE-WEIGHT (SW) MLP

Let $0 < p \leq 1$ and $m \in \mathbb{N}$. We implement each matrix of a static sparse weight FC block with the freezing rate p as follows:

$$x \mapsto x + \phi((M \odot W)\,\mathrm{LN}(x)), \quad x \in \mathbb{R}^m, \tag{S.16}$$

where M is the mask matrix whose $m^2 p$ entries are randomly chosen and set to be one with a probability p and the others are set to be zero with a probability $1 - p$. The mask matrix M is initialized before training and it is frozen during training.

We also consider the SW-MLP consists of sparse weight MLP-blocks as follows:

$$x \mapsto x + \phi\left((M_2 \odot W_2)\phi((M_1 \odot W_2)\,\mathrm{LN}(x))\right), \quad x \in \mathbb{R}^m. \tag{S.17}$$

W_1 and W_2 are weight matrices with hidden features γm, where γ is an expansion factor. M_1, M_2 are mask matrices whose $\gamma m^2 p$ entries are randomly chosen and set to be one with a probability p and the others are set to be zero with a probability $1 - p$.

SW-MLP with L-blocks is composed in the order of the per-patch FC layer, vectorization, L static sparse weight FC blocks (or MLP blocks), and the last classification FC layer.

B. Analysis

B.1. Derivation of Proposition 4.1

For $H = \phi(\phi(WX)V)$, by using $\mathrm{vec}(WXV) = (V^\top \otimes W)\mathrm{vec}(X)$, we have

$$\mathrm{vec}(H) = \phi((V^\top \otimes I_S)\mathrm{vec}(\phi(WX)) \tag{S.18}$$

$$= \phi((V^\top \otimes I_S)\phi((I_C \otimes W)x)). \tag{S.19}$$

Because $J_c^\top (A \otimes B) J_c = (B \otimes A)$ (Magnus & Neudecker, 2019) and any permutation matrix J is commutative with the entry-wise activation function: $J\phi(x) = \phi(Jx)$, we obtain

$$\mathrm{vec}(H) = \phi(J_c^\top (I_S \otimes V^\top)\phi(J_c(I_C \otimes W)x)). \tag{S.20}$$

The skip-less MLP-Mixer (4) is expressed as follows: $Y = \text{Channel-MLP}(\text{Token-MLP}(X)))$, and then

$$u = \phi(J_c(I_C \otimes W_2)\phi((I_C \otimes W_1)x)),$$
$$y = \phi(J_c^\top (I_S \otimes W_4^\top)\phi((I_S \otimes W_3^\top)u)).$$

where $u = \mathrm{vec}(U)$ and $y = \mathrm{vec}(Y)$.

It may be informative that a similar transformation between the matrix and vector is used in a completely different context of deep learning, that is, the Kronecker-factored Approximate Curvature (K-FAC) computation for natural gradient descent (Martens & Grosse, 2015). K-FAC assumes layer-wise preconditioner given by the Kronecker product, that is, $(B \otimes A)^{-1}\mathrm{vec}(\nabla_W Loss(W))$ where A and B correspond to the Gram matrices of the forward and backward signals. This K-FAC gradient can be computed efficiently because it is reduced to a matrix computation of $A^{-1}\nabla_W Loss(W)(B^\top)^{-1}$. Therefore, the trick of formulating a large matrix-vector product for the product among relatively small matrices is common between K-FAC and the aforementioned effective expression.

B.2. Product of Random Permutations

The uniformly distributed random $m \times m$ permutation matrix is given by $J = J_g$ where g is the uniformly distributed random variable on the permutation group S_m of m elements. Then the uniform distribution over S_m is the Haar probability measure, which is translation invariant (see (Folland, 2013) for the detail), that is, $J = J_\sigma J_g$ is also uniformly distributed if σ is S_m-valued uniformly distributed random variables and $g \in S_m$ is constant. Therefore, $J = J_\sigma J_\rho$ is a uniformly distributed random permutation matrix for independent and uniformly distributed random variables σ and ρ on S_m.

B.3. Representation as a PK-family

By (10), one can see that the block of the S-Mixer is

$$U = \text{PK-Layer}_W(X; I, J_c),\ \text{PK-Layer}_{V^\top}(U; I, J_c^\top) \tag{S.21}$$

For MLP-Mixer, the Token-MLP is

$$U = \text{PK-Layer}_{W_2}(\text{PK-Layer}_{W_1}(X; I, J_1); J_1^\top, J_c) \tag{S.22}$$

and Channel-MLP is

$$\text{PK-Layer}_{W_4^\top}(\text{PK-Layer}_{W_3^\top}(U; I, J_2); J_2^\top, J_c^\top) \tag{S.23}$$

for the arbitrary permutation matrices J_1 and J_2.

B.4. Proof of implicit regularization

Here we prove Proposition 3.3 and show a connection between that and (Hoff, 2017). Since each weight V, W are C^2, S^2 dimensional vectors, we only need to show the following Lemma B.1.

Lemma B.1. *Let $m, n \in \mathbb{N}$. We write $||x||_p$ for the L_p norm of any real vector x. Let $f : \mathbb{R}^{mn} \to \mathbb{R}$. Set $h : \mathbb{R}^{mn} \to \mathbb{R}$, $g : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}$ as follows:*

$$\phi(\beta) = f(\beta) + \lambda||\beta||_2, \tag{S.24}$$

$$h(\beta) = f(\beta) + \frac{\lambda}{\sqrt{mn}}||\beta||_1, \tag{S.25}$$

$$g(u, v) = f(u \otimes v) + \lambda(||u||_2^2 + ||v||_2^2)/2. \tag{S.26}$$

Set

$$\Theta_p := \{u \otimes v \mid u \in \mathbb{R}^m, v \in \mathbb{R}^n\}. \tag{S.27}$$

Then it holds that

$$\inf_{u\in\mathbb{R}^m, v\in\mathbb{R}^n} g(u, v) = \inf_{\beta\in\Theta_p} \phi(\beta) \geq \inf_{\beta\in\mathbb{R}^{mn}} h(\beta). \tag{S.28}$$

Proof. Since $||u \otimes v||_2 = ||u||_2||v||_2$, we have

$$\inf_{u\in\mathbb{R}^m, v\in\mathbb{R}^n} g(u, v) = \inf_{\beta\in\Theta_p} \left[f(\beta) + \lambda/2 \inf_{v\in\mathbb{R}^n} (||\beta||_2^2/||v||_2^2 + ||v||_2^2) \right]. \tag{S.29}$$

The inner $\inf$ achieves the maximum when $||v||_2^2 = ||\beta||_2$, and the maximum is $||\beta||_2$. (Note that it is **not squared** L2-norm.) Therefore,

$$\inf_{u\in\mathbb{R}^m, v\in\mathbb{R}^n} g(u, v) = \inf_{\beta\in\Theta_p} (f(\beta) + \lambda||\beta||_2). \tag{S.30}$$

Lastly, by the Hölder's inequality, it hold that $||\beta||_1 \leq ||\beta||_2\sqrt{mn}$, which complete the proof. □

To describe the connection of the implicit L1 regularization in the case of Hadamard product (Hoff, 2017), consider matrix $\mathbf{1}\mathbf{1}^\top$, where $\mathbf{1}$ is the vector with all entries are one. For simplicity, consider the case $S = C$. It is easy to recover the general case by changing the proportional coefficients. Then we have

$$\text{Tr}(W^\top W \otimes \mathbf{1}\mathbf{1}^\top) + \text{Tr}(\mathbf{1}\mathbf{1}^\top \otimes V^\top V) = C^2||W||_F^2 + S^2||V||_F^2 = C^2(||W||_F^2 + ||V||_F^2) \tag{S.31}$$

Put

$$g(W, V) := \mathcal{L}(W \otimes V) + \lambda/2(\|W\|_F^2 + \|V\|_F^2). \tag{S.32}$$

We have

$$g(W,V) = \mathcal{L}(W \otimes V) + \frac{\lambda}{2C^2}(\mathrm{Tr}(W^\top W \otimes \mathbf{1}\mathbf{1}^\top) + \mathrm{Tr}(\mathbf{1}\mathbf{1}^\top \otimes V^\top V)) \tag{S.33}$$

Here consider the vectorization: $u(W) := \mathrm{vec}(W \otimes \mathbf{1}\mathbf{1}^\top)$, $v(V) := \mathrm{vec}(\mathbf{1}\mathbf{1}^\top \otimes V)$. Then

$$u(W) \odot v(V) = \mathrm{vec}(W \otimes V). \tag{S.34}$$

Thus we have

$$\inf_{W,V} g(u,v) = \inf_{W,V} \mathcal{L}(u \circ v) + \frac{\lambda}{2C^2}(\|u\|_2^2 + \|v\|_2^2), \tag{S.35}$$

Because of the domain of u and v, we have

$$\inf_{W,V} g(u,v) \geq \inf_{u,v \in \mathbb{R}^{S^2C^2}} \mathcal{L}(u \circ v) + \frac{\lambda}{2C^2}(\|u\|_2^2 + \|v\|_2^2). \tag{S.36}$$

By Hoff (2017), the right-hand side is equal to

$$\inf_{\beta \in \mathbb{R}^{S^2C^2}} \mathcal{L}(\beta) + \frac{\lambda}{2C^2}\|\beta\|_1. \tag{S.37}$$

Thus the inequality for the Kronecker product has a relationship with (Hoff, 2017) from the vectorization and Kronecker product with $\mathbf{1}\mathbf{1}^\top$.

Let us focus on the normalization factor $1/mn$ and $1/C^2$. In general, there is a trivial relationship between L1 and L2 regularization as follows: For $\theta \in \mathbb{R}^M$,

$$\inf_\theta \mathcal{L}(\theta) + \sqrt{M}\lambda\|\theta\|_2 \geq \inf_\theta \mathcal{L}(\theta) + \lambda\|\theta\|_1, \tag{S.38}$$

this directly follows from the Hölder's inequality. However, in our situation, the parameter space of $W \otimes V$ is a subspace of $\mathbb{R}^{S^2C^2}$, so it is not clear whether the same constant regularization for both L1 and L2 regularization is suitable. At the initialization, we set entries of W, V independently distributed with $\mathcal{N}(0, 1/C)$, thus by the law of large numbers,

$$\|W\|_F^2, \|V\|_F^2 = O(C). \tag{S.39}$$

In the case of dense MLP, we initialize β as $\mathcal{N}(0, 1/SC)$, thus

$$\|\beta\|_1 = O(S^2C^2/\sqrt{SC}) = O(C^3). \tag{S.40}$$

Thus

$$\frac{\|\beta\|_1}{\|W\|_F^2}, \frac{\|\beta\|_1}{\|V\|_F^2} = O(C^2). \tag{S.41}$$

Therefore, the normalizing factor $1/C^2$ in (S.37) matches the scale transformation according to the dimension of parameter spaces. Without the normalizing factor, the L1 regularization term will be C^2 times larger than the loss $\mathcal{L}$.

C. Experimental Setting

C.1. Figure 2

We set the number of blocks of the MLP-Mixer, denoted by L, to 3. Both the Token-mixing block and the Channel-Mixing block are present L times each, resulting in a total of 6 blocks when counted separately. We also set $\gamma = 2$.

For the comparison, the sparse-weight MLP replaces the components within the MLP-Mixer, namely $(I_C \otimes W_1), (I_C \otimes W_2)$ and $(W_3^\top \otimes I_S), (W_4^\top \otimes I_S)$, with the form $M \odot A$. In this context, there's no distinction between the token-mixing and the channel-mixing blocks, leading to a total of 6 blocks.